



TSX and TSXV Level 2 TMX QuantumFeed Business Message Specifications

Document Revision: 3.6.0
Date of Issue: August 10, 2021

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Contact Information

For questions related to TMX QuantumFeed, contact TMX Market Operations at 416.947.4705 or marketops@tmx.com.

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1 Introduction

This document is the *TSX and TSX Venture (TSXV) Level 2 TMX QuantumFeed Business Message Specifications*.

The Level 2 TMX QuantumFeed™ Business messages provide the following information:

- Symbol and market status
- Trades
- Orders and confirmations

1.1 Intended Audience

The intended audience of this specification consists of business analysts and programmer analysts.

1.2 Scope

The scope of this document is limited to describing the *Business messages* disseminated on the TSX and TSXV Level 2 TMX QuantumFeed. The document provides:

- The structure of each type of Business message
- Fields that constitute those messages
- The data types and formats to which the fields comply
- More details on processing the information contained in those fields

To design and implement systems that can receive and process Level 2 TMX QuantumFeed, this document must be used alongside the following documents:

- *TMX QuantumFeed Service Access Guide*
- *TMX eXtreme Message Transfer Protocol Specification*

2 Business Messages Overview

Business messages disseminated on TSX and TSXV Level 2 TMX QuantumFeed contain business-related information; for example: information on booked orders, information on cancelled orders, and so on. A Business message is composed of two sections, a *message header* and the *message body*.

- The **message header** is common to all Business messages and provides information that is required to process the given message.
- The **message body** contains the business-related information.

2.1 Business message types

TSX and TSXV Level 2 TMX QuantumFeed contains the following types of Business messages.

Message Name	Message Type		Description
	ASCII	HEX	
Start-of-Day Messages (Messages disseminated at the start of day)			
Symbol Status	J	0x4A	Symbol Status messages contain information regarding equity, debenture, mutual fund, or trading Instruments for the current trading day for both TSX and TSXV markets. One message is disseminated for each valid symbol at the beginning of each trading day.
Order Book	G	0x47	An Order Book message provides public information for all open non-terms orders in the market at the start of day. One message is disseminated for each valid non-terms order at the beginning of each trading day. Feed clients can use the Order Book message to initialize their order book for the current trading day.
Order Book – Terms	j	0x6A	An Order Book – Terms message provides public information for all open terms orders in the market at the start of day. One message is disseminated for each valid non-terms order at the beginning of each trading day. Feed clients can use the Order Book – Terms message to initialize their order book for the current trading day.

Message Name	Message Type		Description
	ASCII	HEX	
Special Market/Stock State Messages			
Assign COP – Orders	A	0x41	An Assign COP – Orders message specifies the calculated opening price (COP) for a security and the participating orders that are priced at the COP. Each Assign COP – Orders message can contain up to 15 orders priced at COP, and the orders are positional. All the orders reported on a single Assign COP – Orders message always belong to the same market side. Note: This message is produced only if one or more orders are repriced to the COP.
Assign COP – No Orders	B	0x42	An Assign COP – No Order message specifies the calculated opening price for a security when there are no orders that are to be repriced to the COP. Note: This message is produced only if the COP is changing, but no orders are being repriced to the COP.
Assign Limit	C	0x43	An Assign Limit message is disseminated when former better-priced-limit orders are reset to their true limits. Each Assign Limit message can contain up to 15 orders along with their prices, and the orders are positional. All the orders reported on a single Assign Limit message always belong to the same market side.
Intraday messages (Messages that are disseminated intraday)			
Market State Update	E	0x45	A Market State Update message is disseminated each time a notice of a market state change or a trading session change is sent from the Trading Engine.
MOC Imbalance	F	0x46	The MOC Imbalance message is disseminated for MOC-eligible symbols every 10 seconds starting from the beginning of the MOC Imbalance trading session at 3:50 p.m., or as notified by TSX Equities Trading. The message will be disseminated at 4:00 p.m. for closing delayed stocks. The imbalance reference price at that time is the last sale price of the symbol.
Order Booked	P	0x50	An Order Booked message is disseminated in response to a new non-terms order being entered into the trading system.
Order Booked – Terms	m	0x6D	An Order Booked - Terms message is disseminated in response to a new terms order being entered into the trading system.
Order Cancelled	Q	0x51	An Order Cancelled message is disseminated if a non-terms order is cancelled.
Order Cancelled – Terms	n	0x6E	An Order Cancelled – Terms message is disseminated if a terms order is cancelled.

Message Name	Message Type		Description
	ASCII	HEX	
Order Price-Time Assigned	R	0x52	An Order Price-Time Assigned message is disseminated when a new Price or Time is assigned to a non-terms order.
Order Price-Time Assigned – Terms	o	0x6F	An Order Price-Time Assigned – Terms message is disseminated if a new Price or Time is assigned to a terms order.
Stock Status	I	0x49	A Stock Status notification is disseminated in response to a change in stock status from the Trading Engine.
Trade Report	S	0x53	A Trade Report is produced when a trade occurs that has no settlement terms.
Trade Report – Terms	p	0x70	A Trade Report – Terms is produced when a trade occurs that: • Has settlement terms. • Has Non-Resident participation with or without settlement terms.
Trade Cancelled	T	0x54	A Trade Cancelled message is produced when a trade with no settlement terms is cancelled.
Trade Cancelled – Terms	q	0x71	A Trade Cancelled – Terms message is produced when a trade with settlement terms is cancelled.
Trade Correction	U	0x55	A Trade Correction message is produced when a trade with no settlement terms is corrected
Trade Correction – Terms	r	0x72	A Trade Correction – Terms message is produced when a trade with settlement terms is corrected.

2.2 Message data types

Each field that is contained in the Business messages is formatted according to one of the following data types:

Data Type	Description
Alphanumeric	Alphanumeric fields are left-justified ASCII fields (displayable characters from hex 0x20 to hex 0x7E) that are space padded (hex 0x20) on the right and are sized to their corresponding lengths. Example: Consider a field with length <i>10 bytes</i>, data type <i>Alphanumeric</i> and which must contain the information “<i>CASH</i>”. In this scenario, the field will contain the value “<i>CASH</i> ”. The information is left justified and is space padded (“ ”) on the right. The information and the padding (if required) make the field length 10. Default: Space (whole length of the field padded with space)
Binary	Binary fields are unsigned and are Little Endian encoded. Length of the binary fields is one of the following: • 1-Byte Binary • 2-Byte Binary • 4-Byte Binary • 8-Byte Binary Default value: Null (binary value 0)

2.3 Message field definitions

For descriptions of fields that make up the Business messages, refer to “Field Definitions” on page 27.

3 Business Message Formats

This section contains the structure of each Business message disseminated on TSX and TSXV Level 2 TMX QuantumFeed.

Each Business message consists of a message header followed by the message body. For a description of the message header's structure, refer to *TMX eXtreme Message Transfer Protocol Specification*. Only the message body's structure is provided in the following sections.

Note: The size of the header is mentioned for message structure completion purposes.

3.1 Message structure

All Business messages must adhere to the Quantum Message Format, as described in *TMX QuantumFeed Service Access Guide*. Following are the structures of different types of Business messages that are disseminated on the TSX & TSXV Level 2 TMX TM.

3.2 Start-of-Day messages

3.2.1 Symbol Status

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Stock Group	1	1-Byte Binary	
CUSIP	12	Alphanumeric	
Board Lot	2	2-Byte Binary	
Currency	1	Alphanumeric	
Face Value	8	8-Byte Binary	
Last Sale	8	8-Byte Binary	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	
Buy Maximum Quantity	4	4-Byte Binary	
Buy Minimum Quantity	4	4-Byte Binary	
Sell Maximum Quantity	4	4-Byte Binary	
Sell Minimum Quantity	4	4-Byte Binary	
Stock State	2	Alphanumeric	
Test Symbol	1	Alphanumeric	
MOC Eligible	1	Alphanumeric	

Total Message Size = 78 bytes

3.2.2 Order Book

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Broker Number	2	2-Byte Binary	
Order Side	1	Alphanumeric	
Order ID	8	8-Byte Binary	
Price	8	8-Byte Binary	A value of price=0 or 999998 (on the sell or buy side respectively) in the Start-of-Day Order Book message indicates the order is an Oddlot Market-priced GTC order for a halted stock.
Volume	4	4-Byte Binary	
Priority Time Stamp	8	8-Byte Binary	

Total Message Size = 55 bytes

3.2.3 Order Book – Terms

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Broker Number	2	2-Byte Binary	
Order Side	1	Alphanumeric	
Order ID	8	8-Byte Binary	
Price	8	8-Byte Binary	
Volume	4	4-Byte Binary	
Non-Resident	1	Alphanumeric	
Settlement Terms	1	Alphanumeric	
Settlement Date	4	4-Byte Binary	
Priority Time Stamp	8	8-Byte Binary	

Total Message Size = 61 bytes

3.3 Special Market/Stock State Messages

3.3.1 Assign COP – Orders

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Calculated Opening Price	8	8-Byte Binary	
Order Side	1	Alphanumeric	
Broker Number-1	2	2-Byte Binary	
Order ID-1	8	8-Byte Binary	
Broker Number-2	2	2-Byte Binary	
Order ID-2	8	8-Byte Binary	
Broker Number-3	2	2-Byte Binary	
Order ID-3	8	8-Byte Binary	
Broker Number-4	2	2-Byte Binary	
Order ID-4	8	8-Byte Binary	
Broker Number-5	2	2-Byte Binary	
Order ID-5	8	8-Byte Binary	
Broker Number-6	2	2-Byte Binary	
Order ID-6	8	8-Byte Binary	
Broker Number-7	2	2-Byte Binary	
Order ID-7	8	8-Byte Binary	
Broker Number-8	2	2-Byte Binary	
Order ID-8	8	8-Byte Binary	
Broker Number-9	2	2-Byte Binary	
Order ID-9	8	8-Byte Binary	
Broker Number-10	2	2-Byte Binary	
Order ID-10	8	8-Byte Binary	
Broker Number-11	2	2-Byte Binary	
Order ID-11	8	8-Byte Binary	
Broker Number-12	2	2-Byte Binary	
Order ID-12	8	8-Byte Binary	

Field Name	Length (In Bytes)	Data Type	Comments
Broker Number-13	2	2-Byte Binary	
Order ID-13	8	8-Byte Binary	
Broker Number-14	2	2-Byte Binary	
Order ID-14	8	8-Byte Binary	
Broker Number-15	2	2-Byte Binary	
Order ID-15	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 191 bytes

3.3.2 Assign COP – No Orders

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Calculated Opening Price	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 40 bytes

3.3.3 Assign Limit

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Calculated Opening Price	8	8-Byte Binary	
Order Side	1	Alphanumeric	
Broker Number-1	2	2-Byte Binary	
Order ID-1	8	8-Byte Binary	
Price-1	8	8-Byte Binary	
Broker Number-2	2	2-Byte Binary	
Order ID-2	8	8-Byte Binary	
Price-2	8	8-Byte Binary	

Field Name	Length (In Bytes)	Data Type	Comments
Broker Number-3	2	2-Byte Binary	
Order ID-3	8	8-Byte Binary	
Price-3	8	8-Byte Binary	
Broker Number-4	2	2-Byte Binary	
Order ID-4	8	8-Byte Binary	
Price-4	8	8-Byte Binary	
Broker Number-5	2	2-Byte Binary	
Order ID-5	8	8-Byte Binary	
Price-5	8	8-Byte Binary	
Broker Number-6	2	2-Byte Binary	
Order ID-6	8	8-Byte Binary	
Price-6	8	8-Byte Binary	
Broker Number-7	2	2-Byte Binary	
Order ID-7	8	8-Byte Binary	
Price-7	8	8-Byte Binary	
Broker Number-8	2	2-Byte Binary	
Order ID-8	8	8-Byte Binary	
Price-8	8	8-Byte Binary	
Broker Number-9	2	2-Byte Binary	
Order ID-9	8	8-Byte Binary	
Price-9	8	8-Byte Binary	
Broker Number-10	2	2-Byte Binary	
Order ID-10	8	8-Byte Binary	
Price-10	8	8-Byte Binary	
Broker Number-11	2	2-Byte Binary	
Order ID-11	8	8-Byte Binary	
Price-11	8	8-Byte Binary	
Broker Number-12	2	2-Byte Binary	
Order ID-12	8	8-Byte Binary	
Price-12	8	8-Byte Binary	
Broker Number-13	2	2-Byte Binary	
Order ID-13	8	8-Byte Binary	

Field Name	Length (In Bytes)	Data Type	Comments
Price-13	8	8-Byte Binary	
Broker Number-14	2	2-Byte Binary	
Order ID-14	8	8-Byte Binary	
Price-14	8	8-Byte Binary	
Broker Number-15	2	2-Byte Binary	
Order ID-15	8	8-Byte Binary	
Price-15	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 311 bytes

3.4 Intraday messages

3.4.1 Market State Update

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Market State	1	Alphanumeric	
Stock Group	1	1-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 22 bytes

3.4.2 MOC Imbalance

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Imbalance Side	1	Alphanumeric	
Imbalance Volume	4	4-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	
Imbalance Reference Price	8	8-Byte Binary	
Paired Volume	4	4-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.
Market Order Imbalance Volume	4	4-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.
Market Order Imbalance Side	1	Alphanumeric	This field will not be applicable for closing delayed stocks at 4:00p.m.
Near Indicative Closing Price	8	8-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.
Far Indicative Closing Price	8	8-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.

Field Name	Length (In Bytes)	Data Type	Comments
Price Variation	4	4-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.

Total Message Size = 74 bytes

Note: This message is disseminated every 10 seconds per MOC-eligible symbol starting from the beginning of the MOC Imbalance session at 3:50 p.m., or as notified by TSX Equities Trading. The message will also be disseminated at 4:00 p.m. for closing delayed stocks. The imbalance reference price at that time is the last sale price of the symbol.

3.4.3 Order Booked

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Broker Number	2	2-Byte Binary	
Order Side	1	Alphanumeric	
Order ID	8	8-Byte Binary	
Price	8	8-Byte Binary	A value of price=0 or 999998 (on the sell or buy side respectively) in the Intraday Order Booked message indicates an Oddlot Market-priced order entered during the Pre-Open market state, or while a stock is halted.
Volume	4	4-Byte Binary	
Priority Time Stamp	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 63 bytes

3.4.4 Order Booked – Terms

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Broker Number	2	2-Byte Binary	
Order Side	1	Alphanumeric	
Order ID	8	8-Byte Binary	
Price	8	8-Byte Binary	
Volume	4	4-Byte Binary	
Non-Resident	1	Alphanumeric	
Settlement Terms	1	Alphanumeric	
Settlement Date	4	4-Byte Binary	
Priority Time Stamp	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 69 bytes

3.4.5 Order Cancelled

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Broker Number	2	2-Byte Binary	
Order Side	1	Alphanumeric	
Order ID	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 43 bytes

3.4.6 Order Cancelled – Terms

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Broker Number	2	2-Byte Binary	
Order Side	1	Alphanumeric	
Order ID	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 43 bytes

3.4.7 Order Price-Time Assigned

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Broker Number	2	2-Byte Binary	
Order Side	1	Alphanumeric	
Order ID	8	8-Byte Binary	
Price	8	8-Byte Binary	
Volume	4	4-Byte Binary	
Priority Time Stamp	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 63 bytes

3.4.8 Order Price-Time Assigned – Terms

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Broker Number	2	2-Byte Binary	
Order Side	1	Alphanumeric	
Order ID	8	8-Byte Binary	
Price	8	8-Byte Binary	
Volume	4	4-Byte Binary	
Priority Time Stamp	8	8-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 63 bytes

3.4.9 Stock Status

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Comment	40	Alphanumeric	
Stock State	2	Alphanumeric	
Trading System Time Stamp	8	8-Byte Binary	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	

Total Message Size = 76 bytes

3.4.10 Trade Report

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Trade Number	4	4-Byte Binary	
Price	8	8-Byte Binary	
Volume	4	4-Byte Binary	
Buy Broker Number	2	2-Byte Binary	Zero for TSX NAVex™ trades
Buy Order ID	8	8-Byte Binary	
Buy Display Volume	4	4-Byte Binary	
Sell Broker Number	2	2-Byte Binary	Zero for TSX NAVex trades
Sell Order ID	8	8-Byte Binary	
Sell Display Volume	4	4-Byte Binary	
Bypass	1	Alphanumeric	
Trade Time Stamp	4	4-Byte Binary	
Cross Type	1	Alphanumeric	
Trading System Time Stamp	8	8-Byte Binary	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	
Trade Date	4	4-Byte Binary	
Is Mid Only	1	Alphanumeric	
Is Dark	1	Alphanumeric	
Is Conditional	1	Alphanumeric	

Total Message Size = 91 bytes

3.4.11 Trade Report – Terms

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Trade Number	4	4-Byte Binary	
Price	8	8-Byte Binary	

Field Name	Length (In Bytes)	Data Type	Comments
Volume	4	4-Byte Binary	
Buy Broker Number	2	2-Byte Binary	
Buy Order ID	8	8-Byte Binary	
Buy Display Volume	4	4-Byte Binary	
Sell Broker Number	2	2-Byte Binary	
Sell Order ID	8	8-Byte Binary	
Sell Display Volume	4	4-Byte Binary	
Trade Time Stamp	4	4-Byte Binary	
Non-Resident	1	Alphanumeric	
Settlement Terms	1	Alphanumeric	
Settlement Date	4	4-Byte Binary	
Cross Type	1	Alphanumeric	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 87 bytes

3.4.12 Trade Cancelled

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Trade Number	4	4-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	
Trade Date	4	4-Byte Binary	

Total Message Size = 42 bytes

3.4.13 Trade Cancelled – Terms

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Trade Number	4	4-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 36 bytes

3.4.14 Trade Correction

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Trade Number	4	4-Byte Binary	
Price	8	8-Byte Binary	
Volume	4	4-Byte Binary	
Buy Broker Number	2	2-Byte Binary	Zero for TSX NAVex trades
Sell Broker Number	2	2-Byte Binary	Zero for TSX NAVex trades
Initiated By	1	Alphanumeric	
Orig Trade Number	4	4-Byte Binary	
Bypass	1	Alphanumeric	
Trade Time Stamp	4	4-Byte Binary	
Cross Type	1	Alphanumeric	
Trading System Time Stamp	8	8-Byte Binary	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	
Trade Date	4	4-Byte Binary	
Is Mid Only	1	Alphanumeric	
Is Dark	1	Alphanumeric	
Is Conditional	1	Alphanumeric	

Total Message Size = 72 bytes

3.4.15 Trade Correction – Terms

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Trade Number	4	4-Byte Binary	
Price	8	8-Byte Binary	
Volume	4	4-Byte Binary	
Buy Broker Number	2	2-Byte Binary	
Sell Broker Number	2	2-Byte Binary	
Initiated By	1	Alphanumeric	
Orig Trade Number	4	4-Byte Binary	
Trade Time Stamp	4	4-Byte Binary	
Non-Resident	1	Alphanumeric	
Settlement Terms	1	Alphanumeric	
Settlement Date	4	4-Byte Binary	
Cross Type	1	Alphanumeric	
Trading System Time Stamp	8	8-Byte Binary	

Total Message Size = 68 bytes

4 Field Definitions

B

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Board Lot	Contains the board lot size.	Binary (2 Bytes)	No implied decimal places Example: A board lot value of 500 is represented by 11110100 00000001.
Broker Number	An Exchange-assigned public number uniquely identifying a Participating Organization.	Binary (2 Bytes)	No implied decimal places Example: A TSX Broker Number 009 is represented by 00001001 00000000.
Buy Broker Number	An Exchange-assigned public number uniquely identifying a Participating Organization associated with a buy order	Binary (2 Bytes)	No implied decimal places Example: A TSX Broker Number 009 is represented by 00001001 00000000.
Buy Maximum Quantity	The maximum quantity of a mutual fund allowed to purchase.	Binary (4 Bytes)	No implied decimal places Example: A quantity of 2500 is represented by 11000100 00001001 00000000 00000000.
Buy Minimum Quantity	The minimum quantity of a mutual fund required to purchase.	Binary (4 Bytes)	No implied decimal places Example: A quantity of 2500 is represented by 11000100 00001001 00000000 00000000.
Buy Display Volume	Public remaining display volume for the buy order	Binary (4 Bytes)	No implied decimal places Example: A Volume of 2,500 is represented by 11000100 00001001 00000000 00000000.

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Buy Order ID	Public numeric identifier assigned to the buy order by the trading system	Binary (8 Bytes)	No implied decimal places Example: An Order ID 20101010000078654 is represented by 00111110 01100111 10101110 00011000 11000011 01101001 01000111 00000000. The feed client can obtain the Order ID by converting the binary to decimal. The Order ID thus obtained will be of the format YYYYMMDDXXXXXXXXXX, where YYYYMMDD is the business date and XXXXXXXXXX is the sequence number assigned to the order.
Bypass	An indicator for bypass crosses, which can be printed outside of the Protected NBBO and are not subject to cross interference	Alphanumeric (1 Byte)	Valid values are: “Y”: The order is a Bypass. “N”: The order is not a Bypass.

C

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Calculated Opening Price	The price at which orders will trade at the open	Binary (8 Bytes)	6 implied decimal places Example: A Price of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the price.

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Comment	A text field corresponding to a reason code entered when a stock is halted; or, the initiator of a delayed opening on a stock, or when there is a change to the RT/Oddlot trader on a stock. As well, this is a system-produced text field to describe the disabling of the MOC session by TSX Trading Services.	Alphanumeric (40 Bytes)	
Cross Type	Type of crosses originating from a Participating Organization between managed accounts that have the same manager	Alphanumeric (1 Byte)	Valid values are: • "I": Internal • "B": Basis • "C": Contingent • "D": Derivative-Related • "R": Regular • "S": Special Trading session • "V": VWAP – Volume Weighted Average Price • " ": Default, trade is not a cross
Currency	The currency associated with a reported price	Alphanumeric (1 Byte)	Valid values are: • "U": USD • "C": CAD
CUSIP	Clearing and settlement registration number	Alphanumeric (12 Byte)	

F

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Face Value	The face value of a debenture	Binary (8 Bytes)	6 implied decimal places Example: A Face Value of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the face value.
Far Indicative Closing Price	The calculated closing price that will maximize the number of shares matched based on closing interest only (MOC, LOC). This calculation excludes continuous market orders.	Binary (8 Bytes)	See Price.

I

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Imbalance Reference Price	The price that determines if a given limit-priced Market on Close (LOC) order can contribute to the MOC Imbalance. All LOC orders that are priced equal to or more aggressively than this price will contribute to the MOC Imbalance.	Binary (8 Bytes)	See Price.
Imbalance Side	Marker to indicate which side has an imbalance volume for Market on Close	Alphanumeric (1 Byte)	Valid Values are: • "B": Buy Side • "S": Sell Side • " ": No imbalance exists

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Imbalance Volume	Identifies the volume of shares of the imbalance side for Market on Close	Binary (4 Bytes)	No implied decimal places Example: A volume of 2,500 is represented by 11000100 00001001 00000000 00000000.
Initiated By	Indicates which side began the Trade Correction	Alphanumeric (1 Byte)	Valid values are: • “B”: Buy Side • “S”: Sell Side • “C”: Both Buy Side and Sell Side
Is Dark	Indicates the passive side of the trade is dark	Alphanumeric (1 Byte)	“Y” “N”; default is “N”
Is Mid Only	Indicates a dark Contra Midpoint Only Plus to Contra Midpoint Only Plus execution	Alphanumeric (1 Byte)	“Y” “N”; default is “N”
Is Conditional	Indicates a firm-up conditional to firm-up conditional execution	Alphanumeric (1 Byte)	“Y” “N”; default is “N”

L

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Last Sale	Last sale price of a stock	Binary (8 Bytes)	6 implied decimal places Example: A last sale price of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the last sale price.

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Listing Market	The market on which a security is listed.	Alphanumeric (1 Byte)	Valid values are: • "T" = TSX • "V" = TSX Venture • "N" = TSX NAVex • "Q" = Nasdaq • "S" = NYSE • "M" = NYSE MKT • "X" = No market (unlisted securities, posted for trading only)

M

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Market Order Imbalance Volume	Indicates the share Imbalance when considering MOC orders only. Note this will not change from Freeze period as MOC orders are not allowed after this time.	Binary (4 Bytes)	No implied decimal places Example: A volume of 2,500 is represented by 11000100 00001001 00000000 00000000.
Market Order Imbalance Side	Side (buy or sell) of the Market Order Imbalance Volume.	Alphanumeric (1 Byte)	Valid Values are: • "B": Buy Side • "S": Sell Side • " ": No imbalance exists
Market State	Indicates the current market state	Alphanumeric (1 Byte)	For regular stock groups, valid values are: • "P": Pre-open • "O": Opening • "S": Open • "M": MOC Imbalance • "T": MOC Freeze • "A": CCP Determination • "E": Price Movement Extension • "L": Closing • "N": Extended Hours CXLs • "R": Extended Hours Open

Field Name	Description	Data Type and Size	Valid Field Values & Comments
			• "F": Extended Hours Close • "C": Closed For TSX NAVex stock groups, valid values are: • "J": NAVex Closed • "G": NAVex OrderEntry • "H": NAVex OrderEntryClose • "K": NAVex AM Execution • "X": NAVex PM Execution • "Y": NAVex Close
MOC Eligible	Identifies whether the stock is eligible to trade in the Market on Close session each trading day.	Alphanumeric (1 Byte)	Valid values are: • "Y" • "N"

N

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Near Indicative Closing Price	The calculated closing price that will maximize the number of shares matched based on on-close orders (MOC, LOC) and visible continuous market orders. Effectively, this is the price at which the closing print would occur at the time of publication.	Binary (8 Bytes)	• See Price.
Non-Resident	A terms marker indicating that trade participant is not a Canadian resident	Alphanumeric (1 Byte)	Valid values are: • "Y": Participant is not a Canadian resident. "N": Participant is a Canadian resident.

O

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Order ID	Public numeric identifier assigned to an order by the trading system.	Binary (8 Bytes)	No implied decimal places Example: An Order ID of 20101010000078654 is represented by 00111110 01100111 10101110 00011000 11000011 01101001 01000111 00000000. The feed client can obtain the Order ID by converting the binary to decimal. The Order ID thus obtained will be of the format YYYYMMDDXXXXXXXXXX, where YYYYMMDD is the business date and XXXXXXXXXX is the sequence number assigned to the order.
Order Side	Market side of an order	Alphanumeric (1 Byte)	Indicates if an order is a buy order or a sell order Valid values are: • “B” ; Buy • “S”; Sell
Orig Trade Number	Used with trade corrections to reference previously reported executions	Binary (4 Bytes)	No implied decimal places Example: A Trade Number 1000002 is represented by 01000010 01000010 00001111 00000000.

P

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Paired Volume	The number of MOC and LOC shares that are able be matched at the Reference Price.	Binary (4 Bytes)	No implied decimal places Example: A volume of 2,500 Is represented by 11000100 00001001 00000000 00000000.

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Price	The price in a valid currency	Binary (8 Bytes)	6 implied decimal places Example: A Price of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the actual price.
Price Variation	This field indicates the absolute value of the percentage of deviation of the Near Indicative Closing Price from the Reference Price.	Binary (4 Bytes)	2 implied decimal places Example 1: A Price Variation of 25.63 percent is represented by 00000011 00001010 00000000 00000000. Example 2: A Price Variation of 125.63 percent is represented by 00010011 00110001 00000000 00000000. The feed client must convert the binary value to decimal and divide the resulting number by 100 to obtain the actual Price Variation value. Note: The decimal value will be 999.99 if the calculated decimal value exceeds 999.99.
Priority Time Stamp	Time stamp assigned by the Trading Engine to specify time priority of an order. Orders are sequenced in the order book based on symbol, price, and priority time stamp.	Binary (8 Bytes)	Microseconds since the Epoch (00:00:00 UTC, January 1, 1970) Example: A Priority Time Stamp of 2010-10-20 11:12:44.032174 is represented as 10101110 00010000 01011111 11010111 00001101 10010011 00000100 00000000.

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Product Type	The product type for a symbol.	Alphanumeric (1 Byte)	Valid values are: • "B" = Debenture • "E" = Equity • "M" = Mutual Fund • "F" = ETF • "U" = US Equity • "O" = Bond

S

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Sell Broker Number	An Exchange-assigned number uniquely identifying a Participating Organization associated with a sell order	Binary (2 Bytes)	No implied decimal places Example: A TSX Broker Number 009 is represented by 00001001 00000000.
Sell Display Volume	Public remaining volume for the sell order	Binary (4 Bytes)	No implied decimal places Example: A volume of 2500 is represented by 11000100 00001001 00000000 00000000.
Sell Maximum Quantity	The maximum quantity of a mutual fund allowed to sell.	Binary (4 Bytes)	No implied decimal places Example: A quantity of 2500 is represented by 11000100 00001001 00000000 00000000.
Sell Minimum Quantity	The minimum quantity of a mutual fund required to sell.	Binary (4 Bytes)	No implied decimal places Example: A quantity of 2500 is represented by 11000100 00001001 00000000 00000000.

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Sell Order ID	Public numeric identifier assigned to a sell order by the trading system.	Binary (8 Bytes)	No implied decimal places Example: An Order ID 20101010000078654 is represented by 00111110 01100111 10101110 00011000 11000011 01101001 01000111 00000000. The feed client can obtain the Order ID by converting the binary to decimal. The Order ID thus obtained is of the format YYYYMMDDXXXXXXXXXX, where YYYYMMDD is the business date and XXXXXXXXXX is the sequence number assigned to the order.
Settlement Date	Settlement date associated with the order	Binary (4 Bytes)	No implied decimal places Example: A Settlement Date of 20101010 is represented by 10010010 10110111 00110010 00000001. The feed clients can obtain the Settlement Date by converting the binary to decimal. The date thus obtained will be of the format YYYYMMDD.
Settlement Terms	Settlement terms associated with the order	Alphanumeric (1 Byte)	Valid values are: • "C": Cash • "N": NN • "M": MS • "T": CT • "D": If there is a valid Settlement Date associated with the order • " ": No Settlement Terms
Stock Group	An identifier of the stock group	Binary (1 Byte)	No implied decimal places Example: A Stock Group of 3 is represented by 00000011.

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Stock State	The state of the stock	Alphanumeric (2 Bytes)	Valid Values are: • “AR”: AuthorizedDelayed • “IR”: InhibitedDelayed • “AS”: AuthorizedHalted • “IS”: InhibitedHalted • “AE”: Authorized Price Movement Delayed • “IE”: Inhibited Price Movement Delayed • “A”: Authorized • “I”: Inhibited
Symbol	A unique alphanumeric identifier for a security	Alphanumeric (12 Bytes)	

T

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Test Symbol	Indicates the symbol is a test symbol	Alphanumeric (1 Byte)	Valid values are: • “Y” • “N”
Trade Date	Effective date of the trade. Will only be filled when the effective date is different from the current date.	Binary (4 Bytes)	Example: A Trade Date of 20170101 is represented by 01110101 11000101 00110011 00000001. Feed clients can obtain the Trade Date by converting the binary to decimal. The Trade Date format is YYYYMMDD.
Trade Number	Unique identifier assigned to each trade on a per stock basis	Binary (4 Bytes)	No implied decimal places Example: A Trade Number of 1000002 is represented by 01000010 01000010 00001111 00000000.

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Trade Time Stamp	The time at which the trade occurred, manually set when a trade is added	Binary (4 Bytes)	No implied decimal places Example: A Trade Time Stamp of 121010 is represented by 10110010 11011000 00000001 00000000. Feed clients can obtain the Trade Time Stamp by converting the binary to decimal. The Trade Time Stamp is of the format HHMMSS.
Trading System Time Stamp	The time at which the event being reported occurred in the Trading Engine	Binary (8 Bytes)	Nanoseconds since the Epoch (00:00:00 UTC, January 1, 1970) Example: A Trading System Time Stamp of 2015-07-31 10:14:58.496307008 is represented as 01000000 01011111 11001111 11100100 10011100 00001101 11110110 00010011.

V

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Volume	The quantity of shares for an order or fill	Binary (4 Bytes)	No implied decimal places Example: A Volume of 2500 is represented by 11000100 00001001 00000000 00000000.

Appendix A Acronyms

- **CAD**: Canadian dollar
- **CFO**: Change Former Order
- **COP**: Calculated Opening Price
- **CUSIP**: Committee on Uniform Security Identification Procedures
- **MOC**: Market On Close
- **TSX**: Toronto Stock Exchange
- **TSXV**: TSX Venture Exchange
- **USD**: United States dollar
- **UTC**: Coordinated Universal Time
- **VWAP**: Volume Weighted Average Price

Appendix B Revision History

Version	Date	Changes
3.6	2021/4/18	In "Trade Report" and "Trade Correction", added the field "Is Conditional". In "Field Definitions", added "Is Conditional".
3.5.1	2021/04/12	In Intraday messages - "MOC Imbalance" and in "Field Definitions": Changed the field name 'Price Variation Indicator' to 'Price Variation'.
3.5.0	2021/03/25	In Intraday messages - "MOC Imbalance", added clarification about timing of message. In "Field Definitions", added the value "T" to Market State. In "Field Definitions" added the following 6 new fields: 1. Paired Volume 2. Market Order Imbalance Volume 3. Market Order Imbalance Side 4. Near Indicative Closing Price 5. Far Indicative Closing Price 6. Price Variation Indicator
3.4.0	2020/10/20	Changed length of 'Symbol' from 9 to 12 bytes for Business Message Formats and increased corresponding message sizes. In "Field Definitions", updated size of Symbol from 9 to 12 bytes.
3.3.0	2019/05/21	In "Field Definitions", for Listing Market, added the value "X". In "Field Definitions", for Product Type, added the value "O".
3.2.1	2019/07/22	In "Order Book" and "Order Booked", updated the comment for the field "Price". In "Field Definitions", updated the definition of "Is Mid Only"
3.2.0	2019/04/16	In "Trade Report" and "Trade Correction", added the field "Is Dark". In "Field Definitions", added "Is Dark".
3.1.0	2018/01/31	In "Trade Report" and "Trade Correction", added the field "Is Mid Only". In "Field Definitions", added "Is Mid Only".
3.0.0	2017/05/08	In "Business message types", updated the description of Trade Report – Terms. In "Field Definitions", updated the description of Bypass. In "Field Definitions", for Cross Type, added the values "D" and " ". In "Field Definitions", for Settlement Terms, added the value " ".
2.50	2017/02/07	In "Field Definitions", added "Q" = Nasdaq, "S" = NYSE, "M" = NYSE MKT to Listing Market. In "Field Definitions", added "U" = US Equity to Product Type.
2.40	2016/11/29	In "Symbol Status", added the field MOC Eligible.

Version	Date	Changes
		In "Trade Report", "Trade Cancelled", and "Trade Correction", added the field Trade Date. In "Field Definitions", changed Market State value "X – NAVex Execution" to "K – NAVex AM Execution" and "X – NAVex PM Execution". In "Field Definitions", added MOC Eligible and Trade Date.
2.30	2016/08/03	In "Business message types", for MOC Imbalance, added "and again for closing delayed stocks" In "Symbol Status", added the Stock State and Test Symbol fields. In "Field Definitions", added the value "R" to Cross Type. In "Field Definitions", added the value "F" to Product Type. In "Field Definitions", removed the following values from Stock State: "AG" Authorized Frozen, "IG" Inhibited Frozen, "AF" Authorized Price Movement Frozen, and "IF" Inhibited Price Movement Frozen. In "Field Definitions", added the Test Symbol field.
2.20	2016/03/30	In Business Message Types, added mutual fund to Stock Status message description. In the Market State Message, added TSX NAVex market states. In the Trade Report and Trade Correction messages, noted that Buy Broker Number and Sell Broker Number will be zero for TSX NAVex trades. Added Listing Market and Product Type to the Symbol Status, Stock Status, Trade Report, Trade Cancelled, and Trade Correction messages. Added Buy Maximum Quantity, Buy Minimum Quantity, Sell Maximum Quantity, and Sell Minimum Quantity to the Symbol Status message. Added Buy Maximum Quantity, Buy Minimum Quantity, Listing Market, Product Type, Sell Maximum Quantity, and Sell Minimum Quantity to Field Definitions.
2.13	2015/10/30	Added comment to Price field of Order Book and Order Booked messages.
2.12	2015/08/11	Modified the definition of Trading System Time Stamp to indicate that it supports nanoseconds.
2.11	2015/04/30	Changed the Field Definition for Cross Type to remove 'TSX only' from the flags.
2.10	2014/06/26	Added section 2.4, "Message version". Added Imbalance Reference Price to MOC Imbalance message and Field Definitions.
2.02	2013/07/12	Added Trading System Time Stamp field to Order Booked and Order Booked – Terms messages.
2.01	2013/04/10	Branding and minor editorial corrections.
2.0	2010/11/16	Initial release.

Version	Date	Changes
1.0	2010/11/12	Internal release.



TMX Market Operations phone: 416.947.4705
300 – 100 Adelaide St. W. email: marketops@tmx.com
Toronto, ON
M5H 1S3